

2DMamba: Efficient State Space Model for Image Representation with Applications on Giga-Pixel Whole Slide Image Classification

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Abstract

Efficiently modeling large 2D contexts is essential for various fields including Giga-Pixel Whole Slide Imaging (WSI) and remote sensing. Transformer-based models offer high parallelism but face challenges due to their quadratic complexity for handling long sequences. Recently, Mamba introduced a selective State Space Model (SSM) with linear complexity and high parallelism, enabling effective and efficient modeling of wide context in 1D sequences. However, extending Mamba to vision tasks, which inherently involve 2D structures, results in spatial discrepancies due to the limitations of 1D sequence processing. On the other hand, current 2D SSMs inherently model 2D structures but they suffer from prohibitively slow computation due to the lack of efficient parallel algorithms. In this work, we propose 2DMamba, a novel 2D selective SSM framework that incorporates the 2D spatial structure of images into Mamba, with a highly optimized hardware-aware operator, adopting both spatial continuity and computational efficiency. We validate the versatility of our approach on both WSIs and natural images. Extensive experiments on 10 public datasets for WSI classification and survival analysis show that 2DMamba improves up to 2.48% in AUC, 3.11% in F1 score, 2.47% in accuracy and 5.52% in C-index. Additionally, integrating our method with VMamba for natural imaging yields 0.5 to 0.7 improvements in mIoU on the ADE20k semantic segmentation dataset, and 0.2% accuracy improvement on ImageNet-1K classification dataset. Our code is available at <https://github.com/AtlasAnalyticsLab/2DMamba>.

1. Introduction

Efficient understanding of large contexts over a 2D visual domain is crucial across fields such as medical imaging

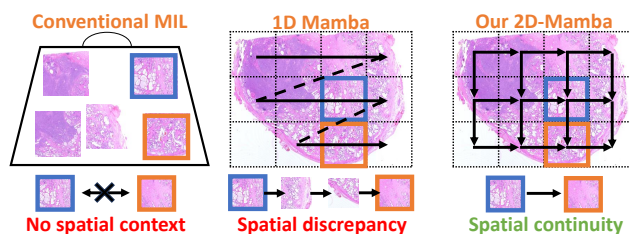


Figure 1. **Left: Conventional MIL** Bagging of patches adopts no spatial context. **Middle: 1D Mamba-based methods** flatten a WSI into a 1D sequence and lose the 2D structure. The adjacent blue and orange patches are far away in the sequence. We call this “spatial discrepancy”. **Right: 2DMamba** processes a WSI in a 2D manner, preserving 2D structure. The blue and orange patches maintains adjacent in the sequence. We call it “spatial continuity”.

and remote sensing [7, 8, 18]. While recurrent neural networks (RNNs) [38] can model wide context in long sequences, their sequential nature limits parallelism, making them unable to fully utilize GPUs. As a remedy, Transformers [12, 43], possessing a high capacity of parallelism, became the mainstream to model long sequences, albeit with quadratic complexity. As a solution, Mamba [10, 15], benefiting from both linear-time complexity and parallelism, has emerged as a promising avenue. Mamba is a State Space Model (SSM), a mathematical framework used in control theory to capture dynamic interactions between state variables [14, 22, 37]. It introduces a selective mechanism that enhances the flexibility of SSMs, allowing them to capture essential information and ignore irrelevant context.

Mamba was proposed for language modeling by only processing 1D sequences and extended to vision domain [26, 49]. Due to the 2D nature of vision tasks, Mamba-based approaches for natural images have attempted to incorporate 2D image structures by adopting various formulations for flattening 2D images into 1D sequences or by scanning images in multiple directions simultaneously

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[26, 36, 45, 49]. However, these methods **all** flatten 2D images into 1D sequences, which inevitably leads to the loss of spatial structure in at least one direction. This limitation persists regardless of the specific flattening strategy employed, resulting in suboptimal performance. We call this issue as “*spatial discrepancy*” illustrated in Fig. 1.

One alternative solution is 2D SSMs to maintain spatial continuity of 2D structures [2, 13]. However, unlike the highly parallel Mamba architecture, achieving parallel implementation for these methods still remains a big challenge. As a consequence, similar to traditional RNNs, these methods experience very slow computation, making them almost impractical. Moreover, they lack Mamba’s selective mechanism, resulting in suboptimal performance.

Beyond applications in general vision tasks, Mamba finds a great potential in computational pathology, particularly for the classification of Giga-pixel Whole Slide Images (WSIs), known as the gold standard for cancer diagnosis [19, 31–33, 39, 51]. WSIs are high-resolution images of tissue samples, often reaching up to $100,000 \times 100,000$ pixels at 40x magnification, making them extremely large and rich in spatial detail. Due to their enormous size, WSIs are typically analyzed in a Multiple Instance Learning (MIL) manner: conventional bag-based MIL methods convert a WSI into a “bag” of instances (patches) that are usually aggregated independently, neglecting the spatial awareness among patches [21, 29, 48]. In contrast, Mamba-based methods treat the WSI as a sequence of patches [13, 46], enabling more effective information aggregation and potentially enhancing diagnostic insights. However, they still flatten 2D images into 1D sequences, and *spatial discrepancies* persist, as illustrated in Fig. 1. Given that cells interact with each other in a coordinated manner across all directions, scanning in multiple directions [26, 40] does not accurately model the complexity of cell-to-cell interactions.

We propose a novel framework **2DMamba** to overcome the limitations posed by the 1D nature of Mamba and the sequential nature of 2D SSMs. In summary,

- We propose a *2D selective State Space Model architecture* which directly scans a 2D image without first flattening it into a 1D sequence. It maintains the 2D structure of images and we call this “*spatial continuity*” (Fig. 1).
- We propose a novel *hardware-aware 2D selective scan operator* to extend the 1D Mamba parallelism into 2D.
- We validate the versatility of our architecture by implementing it on two very different domains, Giga-pixel WSIs for MIL aggregation and 224×224 natural images.

To our best knowledge, 2DMamba is the **first** intrinsic 2D Mamba method with an efficient parallel algorithm. Extensive experiments on 10 public datasets for WSI classification and survival analysis show that our method achieves a relative improvement of up to 2.48% better AUC, 3.11% better F1, 2.47% better Accuracy, and 5.52% better C-

index. We also integrate our scanning approach into the SOTA method, VMamba [26]. We outperform the SOTA method by 0.5 to 0.7 in mIoU on the ADE20k semantic segmentation dataset and max the SOTA method by 0.2% in accuracy on the ImageNet-1K classification dataset.

2. Related work

State Space Model (SSM). SSM [23] is an effective sequence model that represents systems evolving over time by defining hidden states and their transitions, which makes it particularly useful for capturing dynamic temporal behavior in sequential data. Gu et al. [17] unified RNNs, temporal convolutions, and neural differential equations with a linear state-space layer and demonstrated the potential of SSM-based models with the HiPPO initialization. S4 [16] proposed to normalize the parameter matrices into a diagonal structure. 2D-SSM [2] adopted Roesser’s 2D-SSM recursion [24] and applied it to 2D images. However, prior to Mamba, all these SSM methods suffered from slow training speed as the sequential dependency of states makes an efficient parallel algorithm very difficult.

Mamba. To accelerate SSM methods, Mamba [15] incorporated a selective mechanism that makes the model parameters input-dependent and eliminates long dependencies by forgetting less relevant states. It also introduced a hardware-aware algorithm that drastically accelerates state computation. It was originally applied to language tasks and Vim [49] introduced a Vision Mamba block that uses two independent selective SSMs for bidirectional aggregation of information in the vision domain. PlainMamba [45] used a 4-directional selective scan and adopted a more spatially continuous scan path. Similarly, VMamba [26] and GroupMamba [40] also utilized this 4-directional scan in a hierarchical network and optimized the network structure. However, the current formulations of these Mamba-based models are still limited to 1D.

Application of MIL in WSI classification. MIL methods are the mainstream on WSI classifications. It aggregates embedded features from a WSI for slide-level representation. AB-MIL [21] introduced an attention-based aggregation, where the attention values were learned by a neural network. Based on that, CLAM [29] proposed a multi-branch pooling mechanism to improve the performance. DSMIL [25] employed multi-scale patch features in a dual-stream architecture. TransMIL [41] introduced multi-head self-attention layers to capture both morphological and spatial relationships and used nyström Attention [44] to alleviate the quadratic complexity of self-attention. DTFD-MIL [47] introduced a double-tier MIL framework by incorporating pseudo-bags. Recently, S4-MIL [13] and MambaMIL [46] used Mamba for better capturing the information in long patch sequences. However, these works still fail to fully utilize the 2D spatial information of a WSI.

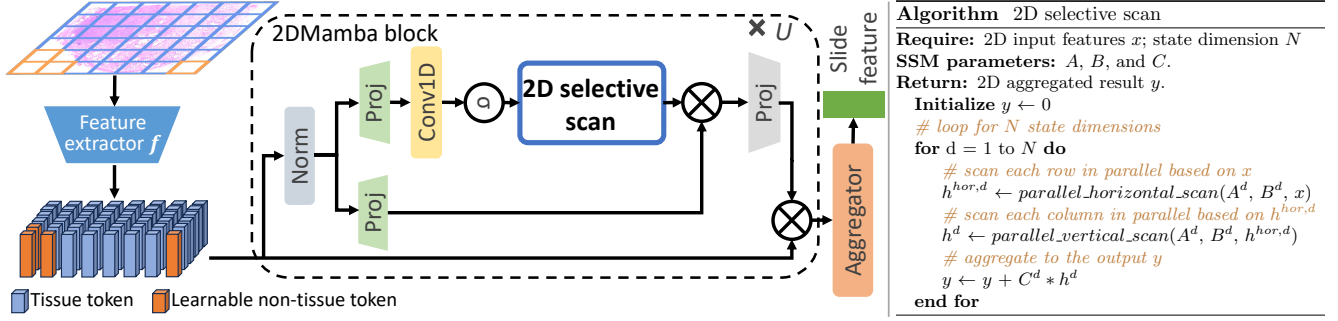


Figure 2. **Left:** The overall architecture of 2DMambaMIL for WSI representation. An input WSI is first tiled into patches and these patches are embedded by a feature extractor into a 2D features map. Non-tissue regions are padded with the learnable token to maintain the 2D spatial relationships. The 2D feature map is then fed to U layers of 2D-Mamba blocks, where the key difference, compared with vanilla Mamba block, is our **2D selective scan** module. **Right:** Our 2D selective scan algorithm. It performs parallel horizontal scan and parallel vertical scan for each state dimension d independently. Parameter C then aggregates N state dimensions into a single dimension output y .

3. Method

We present our 2DMamba designed for effectiveness and efficiency, and an associated framework for WSI representation: 2DMambaMIL.

3.1. SSM in Mamba and 1D selective scan

We revisit SSM, a mathematical model used to capture the behavior of dynamic systems. SSMs are designed as a *function-to-function* for continuous systems and after discretization it becomes a *sequence-to-sequence* model:

$$h_t^d = \bar{A}^d h_{t-1} + \bar{B}^d x_t^d \quad (1)$$

$$y_t = C h_t = \sum_{d=1}^N C^d h_t^d, \quad (2)$$

where h_t^d is the latent state at time t , y_t is the output, and $d \in \{1, 2, \dots, N\}$ is the state dimension. The parameters $\bar{A}^d$ and $\bar{B}^d$ are time-invariant, making them non-adaptive to the input. This design limits the context-aware ability of SSMs to handle long sequence inputs.

The vanilla Mamba block [15] introduces a selective mechanism to allow the SSM to dynamically adapt to the input context. This aggregates important input into the hidden state while unimportant input can be ignored. Mathematically, the parameters are formulated as functions of the input x_t :

$$\begin{aligned} \bar{A}_t^d &= \exp(\Delta_t A^d), & \bar{B}_t^d &= \Delta_t B^d(x_t), \\ C_t^d &= C^d(x_t), & \Delta_t &= \text{softplus}(\Delta(x_t)), \end{aligned} \quad (3)$$

where Δ , B^d , and C^d are learnable linear functions of x_t . Δ_t represents the time step of the discretization. The selective mechanism in the Mamba block is commonly referred to as a selective scan. For better distinguishing with our 2D method, we refer to this scan as *1D selective scan* due to its 1D scanning process.

3.2. Architecture of 2DMambaMIL

The overall architecture of 2DMambaMIL is illustrated in Fig.2. The model includes: U layers of 2DMamba blocks and an aggregator. The first component is the stack of U 2DMamba layers. We utilize the design of the original Mamba block [15] and replace the original 1D selective scan with our 2D variant. The second component is an aggregator, which is an attention based module with two linear projections, producing a slide feature.

Our 2DMambaMIL first tiles an input WSI into patches $\{X_{i,j}\}$ with $i \in \{1, 2, \dots, H\}$ and $j \in \{1, 2, \dots, W\}$. H and W denote the number of the tiled patches along the height and width dimensions, respectively. These patches are then embedded differently based on their types. Tissue patches are embedded using a pre-trained pathology feature extractor f . Additionally, we propose using a learnable token p to represent non-tissue patches padded to obtain a 2D rectangle feature map. This allows the model to learn the proper representation of non-tissue regions during training. Formally, the WSI is transformed into a feature map x with a rectangle shape (H, W) :

$$x_{i,j} = \begin{cases} f(X_{i,j}) & \text{if } X_{i,j} \text{ is a tissue patch} \\ p & \text{otherwise} \end{cases} \quad (4)$$

3.3. 2D selective SSM architecture.

We detail 2D selective SSM architecture. The key component of 2DMamba is the *2D selective scan* operation. In contrast to vanilla mamba which aggregates information from a flattened 1D sequence, 2DMamba aggregates both geometric and semantic information directly from a 2D feature map. Particularly, 2DMamba conducts both horizontal and vertical scans in parallel. For simplicity, we omit the state dimension superscript d in this section. The parameters of the 2D selective scan remain the same as 1D in

Eq. (3), with the subscript being (i, j) to index 2D inputs instead of t . We reuse $x_{i,j}$ to represent the input of the 2D selective scan after normalization, projection, and convolution layers in Fig. 2.

We formulate 2D scanning in a manner similar to the vanilla Mamba to maintain efficient parallelism. As shown in Fig. 2, we first conduct a horizontal scan on each row independently, equivalent to applying a 1D selective scan to each row. Specifically, the state $h_{i,j}^{\text{hor}}$ obtained during the horizontal scan is:

$$h_{i,j}^{\text{hor}} = \bar{A}_{i,j} h_{i,j-1}^{\text{hor}} + \bar{B}_{i,j} x_{i,j}. \quad (5)$$

Note that, for the first column, we assume $h_{i,0}^{\text{hor}} = 0$, and thus $h_{i,1}^{\text{hor}} = \bar{B}_{i,1} x_{i,1}$. Two parameters $\bar{A}_{i,j}$ and $\bar{B}_{i,j}$, which depend on $x_{i,j}$, regulates the information of previous states $h_{i,j-1}$ and current input $x_{i,j}$.

After the horizontal scan, we apply our vertical scan on each column of $h_{i,j}^{\text{hor}}$ independently. Compared with horizontal scan, we replace $\bar{B}_{i,j} x_{i,j}$ with the result $h_{i,j}^{\text{hor}}$ obtained from the horizontal scan.

$$h_{i,j} = \bar{A}_{i,j} h_{i-1,j} + h_{i,j}^{\text{hor}}. \quad (6)$$

Note that for the first row $h_{1,j} = h_{1,j}^{\text{hor}}$ by assuming $h_{0,j}^{\text{hor}} = 0$. We reuse the same $\bar{A}_{i,j}$ for the vertical scan.

If we omit the subscripts of $\bar{A}$ and $\bar{B}$, and expand Eqs. (5) (6), the hidden state $h_{i,j}$ can be formulated as the following equation (detailed derivations in Supp. C):

$$h_{i,j} = \sum_{i' \leq i} \sum_{j' \leq j} \bar{A}^{i-i'+j-j'} \bar{B} x_{i',j'}, \quad (7)$$

After two scans, the output y is aggregated from h by the parameter C similar to 1D-Mamba: $y_{i,j} = C h_{i,j}$. For each location (i, j) , the aggregation information is obtained from its upper left locations.

By doing so, 2DMamba aggregates information without *spatial discrepancy*. In comparison, the hidden state of the vanilla Mamba on a flattened image is given by $h_i^{1D} = \sum_{i' \leq i} \bar{A}^{i-i'} \bar{B} x_{i'}$ where i denotes the 1D index. The order $i-i'$ represents the distance between i, i' in a flattened sequence, where a higher order (larger distance) may lead to forgetting [42]. This mathematically encapsulates the concept of “*spatial discrepancy*”. In contrast, 2DMamba achieves the formulation in Eq.(7), where the order $i-i'+j-j'$ corresponds to the Manhattan distance between (i', j') and (i, j) , thereby preserving the 2D structure. This distance represents a path from (i', j') to (i, j) that moves horizontally to the right and then vertically downward. This mathematically encapsulates the concept of “*spatial continuity*”. For instance, the last hidden state for a 3x3 feature map can be expressed as:

$$\begin{aligned} h_{3,3}^{1D} &= \bar{A}^8 \bar{B} x_{1,1} + \bar{A}^7 \bar{B} x_{1,2} + \bar{A}^6 \bar{B} x_{1,3} + \bar{A}^5 \bar{B} x_{2,1} + \bar{A}^4 \bar{B} x_{2,2} + \bar{A}^3 \bar{B} x_{2,3} + \bar{A}^2 \bar{B} x_{3,1} + \bar{A}^1 \bar{B} x_{3,2} + \bar{A}^0 \bar{B} x_{3,3} \\ h_{3,3}^{2D} &= \bar{A}^4 \bar{B} x_{1,1} + \bar{A}^3 \bar{B} x_{1,2} + \bar{A}^2 \bar{B} x_{1,3} + \bar{A}^3 \bar{B} x_{2,1} + \bar{A}^2 \bar{B} x_{2,2} + \bar{A}^1 \bar{B} x_{2,3} + \bar{A}^2 \bar{B} x_{3,1} + \bar{A}^1 \bar{B} x_{3,2} + \bar{A}^0 \bar{B} x_{3,3} \end{aligned} \quad (8)$$

where a much larger order term $\bar{A}^8$ of $x_{1,1}$ in the 1D case (compared to $\bar{A}^4$ in 2D case) results in much more forgetting and a loss of the 2D structure information. Notably, *spatial discrepancy* becomes particularly problematic in WSIs, as it leads to significantly larger order terms (e.g. $\bar{A}^{200}$) due to the large size of WSIs.

3.4. Hardware-Aware 2D Selective Scan

We present our *hardware-aware scanning operator* that accelerates 2D selective scans. First, we revisit the GPU memory hierarchy and analyze the major challenges for 2D selective scans. Then, we present our novel operator in detail.

GPU memory hierarchy. Fig. 3 (d) illustrates the memory hierarchy of modern GPUs. The **green** area represents off-chip GPU memory, with low speed and high capacity. Here, it is referred to as *high bandwidth memory* (HBM). The **orange** area denotes on-chip memory, with high speed but low capacity, and is referred to as SRAM. In GPU algorithms, data is transferred from HBM to SRAM for computation, and the results are stored back to HBM to vacate SRAM for succeeding computation. Memory transfers are expensive. Therefore, instead of computation, many GPU algorithms [9, 11] are bounded by memory. Mamba’s selective scan [15] is also *memory-bounded*.

Mamba’s 1D selective scan. The vanilla Mamba is fast as the GPU memory follows a hierarchy by 1D tiling and caching. As shown in Fig. 3 (a), a long sequence in HBM is divided into smaller tiles. Each tile is loaded into SRAM, scanned across N independent state dimensions, aggregated into a single output by rules specified in Eq. (2), and stored back to HBM. The intermediate results of the N state dimensions are *not materialized on HBM*, and will be recomputed during back-propagation. The overall memory access complexity is $\mathcal{O}(L)$, where L denotes the sequence length.

Naive 2D selective scan. It is not trivial to extend 1D Mamba scans to 2D. As illustrated in Fig. 3 (b), a naive extension to 1D Mamba will scan a 2D feature map in two steps. First, the feature map is tiled into H rows for row-wise 1D Mamba scans. Next, the succeeding vertical scans must be applied independently to each column, where each column has N independent state dimensions. Therefore, the horizontal scanner *must materialize* N intermediate feature maps on HBM. Each feature map is then tiled into W columns for column-wise 1D Mamba scans. Its memory access complexity is $\mathcal{O}(NHW) = \mathcal{O}(NL)$, which, as demonstrated in Table 3, results in low throughput and high memory consumption.

Hardware-aware 2D selective scan. The proposed hardware-aware 2D selective scan operator, which is illustrated in Fig. 3 (c), optimizes memory transactions by 2D tiling and caching. Instead of tiling by rows or by columns, we divide the feature map into a 2D grid. At each step, we only load a small submatrix into SRAM. Then we con-

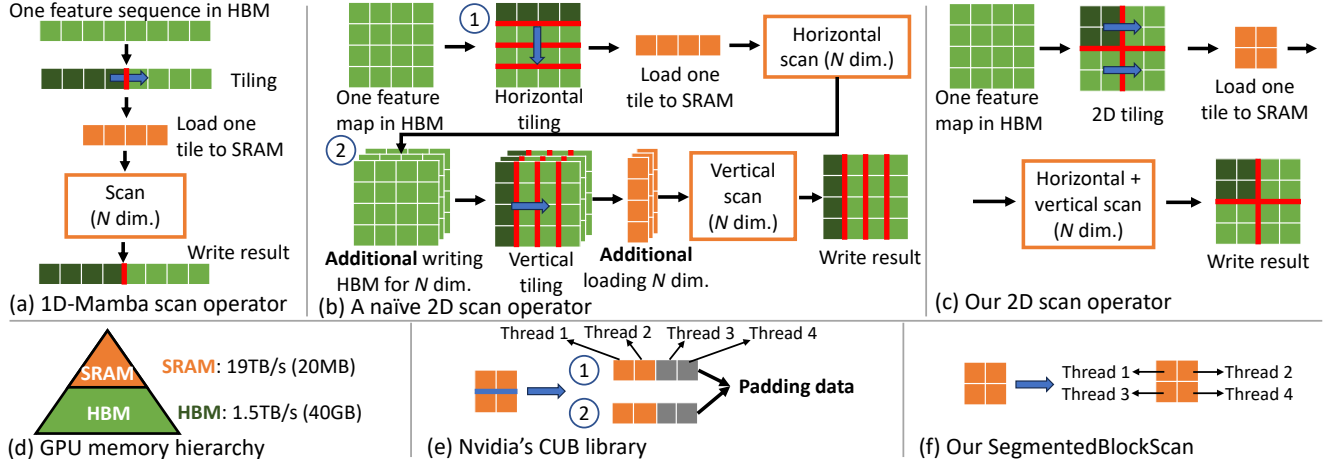


Figure 3. Our hardware-aware 2D selective scan operator with efficient caching mechanism and high parallelism. **Orange** color represents operations on SRAM and **green** color represents those on HBM. (a) The **1D Mamba scan operator** intakes a flattened sequence on HBM. It tiles the input into sub-sequences. Each sub-sequence is loaded from HBM to SRAM, scanned and reduced across N intermediate dimensions, and then written back to HBM. The total memory access complexity is $\mathcal{O}(L)$. (b) A **Naïve 2D scan operator** tiles the 2D feature map by rows and columns, and performs 1D Mamba scans on each row, column, and on N independent state dimensions. This will explicitly instantiate N intermediate feature maps on HBM, resulting in a memory access complexity of $\mathcal{O}(NL)$. (c) **Our 2D scan operator** tiles the feature map into 2D grids and scans each grid in 2 directions. Intermediate features are reduced inside each tile, only the aggregated result is stored back to HBM. The memory complexity is $\mathcal{O}(L)$. (d) **GPU memory hierarchy**: SRAMs are small but fast; HBMs have large capacities but are slow. (e) **NVIDIA’s CUB BlockScan** only supports 1D sequences, with sizes of multiples of 32. Scanning a two-row grid requires two sequential kernel launches and padding elements. (f) **Our SegmentedBlockScan** enables scanning multiple rows and columns in parallel. It reduces the amount of memory transactions and padding data.

duct horizontal and vertical scans for N independent state dimensions, and write the aggregated output back to HBM. This *avoids the explicit materialization* of the state dimensions, and maintains an overall memory access complexity of $\mathcal{O}(HW) = \mathcal{O}(L)$, equivalent to the vanilla Mamba.

Moreover, vanilla Mamba employs NVIDIA’s CUB library [34] for 1D parallel scans. However, as illustrated in Fig. 3 (e), CUB’s BlockScan algorithm only supports full-sequence scanning. Thus, it requires multiple scans for a multi-row feature map. Moreover, for a 2D feature map, CUB BlockScan requires both its height H and width W to be multiples of 32, where 32 is the smallest thread schedule granularity for NVIDIA GPUs. Therefore, small feature maps must be padded before computation, leading to inefficiencies. For instance, a typical 14×14 feature map will require 18 padding elements per row and column, wasting as much as 56% of computation. To resolve this limitation, we introduce the SegmentedBlockScan algorithm, which is illustrated in Fig. 3 (f). It distributes GPU threads across both rows and columns, only requiring $H \times W$ to be a multiple of 32. This enables simultaneous multi-row/column scanning and significantly reduces the padding requirements for small feature maps. For instance, regarding the same 14×14 feature map, our method requires only 2 padding elements per row and column. Detailed algorithm and implementations can be found in Supp. B.

4. Experiments

4.1. Dataset

We assess 2DMambaMIL on 5 public pathology classification datasets, TCGA-BRCA [1], BRACS [3], PANDA [4], DHMC [50], TCGA-NSCLC and 5 public survival datasets, TCGA-(KIRC, KIRP, LUAD, STAD, UCEC). These datasets cover a variety of organs including breast, prostate, lung, kidney, stomach, and uterine. The number of slides ranges from 261 to 10614 and we use 20x magnification for all these datasets. Details of the datasets are listed in the Supp. D. Following [26], we evaluate our 2DMamba on two natural image datasets of ImageNet-1K classification and ADE20K semantic segmentation.

4.2. Results

WSI Classification. We compare 2DMambaMIL with eight other SOTA MILs on five WSI classification datasets. The baselines include ABMIL [21], CLAM [29], DSMIL [25], DTFDMIL [47], TransMIL [41], S4-MIL [13], MambaMIL [46] and SRMambaMIL [46]. The first four MILs are attention-based, TransMIL is Transformer-based, and the last three are 1D SSM-based MILs. We utilize three metrics to evaluate the WSI classification performance: accuracy (Acc), F1 score ($F1$), and area under the curve (AUC). Table 1 shows that our 2DMambaMIL surpasses all

Method	BRACS			DHMC			PANDA			TCGA-NSCLC			TCGA-BRCA		
	Acc	F1	AUC	Acc	F1	AUC	Acc	F1	AUC	Acc	F1	AUC	Acc	F1	AUC
AB-MIL	0.7057	0.6015	0.8939	0.8684	0.7774	0.9695	0.4883	0.4269	0.7797	0.8758	0.8756	0.9572	0.9292	0.8893	0.9747
DSMIL	0.6759	0.5618	0.8618	0.8711	0.7934	0.9583	0.4633	0.3847	0.7660	0.8782	0.8780	0.9567	0.9375	0.8961	0.9770
CLAM	0.7103	0.6014	0.9016	0.8711	0.7909	0.9727	0.4802	0.4224	0.7820	0.8804	0.8803	0.9536	0.9333	0.8960	0.9753
DTFD-MIL	0.7012	0.6131	0.8787	0.8711	0.7704	0.9521	0.4704	0.3853	0.7665	0.8736	0.8732	0.9559	0.9271	0.8809	0.9633
TransMIL	0.6919	0.6063	0.8759	0.8067	0.7136	0.9466	0.4636	0.3970	0.7728	0.8850	0.8845	0.9626	0.9375	0.9028	0.9763
S4-MIL	0.6621	0.5904	0.8457	0.8644	0.7847	0.9284	0.5047	0.4486	0.7986	0.8851	0.8849	0.9571	0.9458	0.9154	0.9770
MambaMIL	0.7379	0.6832	0.8883	0.8550	0.7789	0.9661	0.4679	0.4216	0.7781	0.8758	0.8756	0.9582	0.9333	0.8939	0.9657
SRMambaMIL	0.7379	0.6789	0.8915	0.8590	0.7735	0.9639	0.4711	0.4209	0.7776	0.8850	0.8849	0.9592	0.9313	0.8900	0.9657
2DMambaMIL	0.7517	0.7045	0.8964	0.8926	0.8027	0.9468	0.5075	0.4562	0.8184	0.8851	0.8850	0.9618	0.9458	0.9156	0.9782

Table 1. The comparison of accuracy (Acc), F1 and AUC on five **WSI classification** datasets. We conducted each experiment five times using five different random seeds and reported their mean. The highest metrics are marked as **bold**.

current SOTA methods across multiple datasets, indicating our strong generalization ability. Compared with the best-performing non-Mamba method, we achieve significant improvements of up to 5.83% in accuracy, 14.90% in F1 score, and 4.65% in AUC. 2DMambaMIL also outperforms SSM-based methods by up to 3.26% in accuracy, 3.11% in F1 score, and 2.48% in AUC, showing the benefit of preserving spatial continuity in WSIs.

WSI Survival Analysis. We further compare 2DMambaMIL with all eight MILs on five WSI survival datasets. We assess the performance using the concordance index (C-index), which evaluates how well a survival model ranks patients with their survival time compared to the actual survival outcomes. As shown in Table 2, 2DMambaMIL consistently achieves the highest C-index scores across all datasets, indicating superior predictive performance. Specifically, 2DMambaMIL achieves a relative improvement of 0.6%, 1.2%, 5.5%, 2.9%, and 1.0% on C-index compared with the best-performing baseline on KIRC, KIRP, LUAD, STAD, and UCEC, respectively.

Method	KIRC	KIRP	LUAD	STAD	UCEC
ABMIL	0.7051	0.7824	0.6157	0.6119	0.7243
DSMIL	0.6240	0.7122	0.6114	0.6010	0.6324
CLAM	0.5723	0.7197	0.5874	0.5883	0.6312
DTFD-MIL	0.7271	0.7933	0.6020	0.6168	0.7462
TransMIL	0.6944	0.7317	0.6139	0.5978	0.6997
S4-MIL	0.7232	0.7905	0.5945	0.6001	0.7459
MambaMIL	0.7096	0.7822	0.5952	0.6244	0.7419
SRMambaMIL	0.7178	0.7424	0.5876	0.6130	0.7398
2DMambaMIL	0.7311	0.8027	0.6198	0.6428	0.7536

Table 2. The comparison of C-Index on five **survival analysis** datasets. We performed 5-fold cross-validation for all experiments. The highest metrics are **bold**.

Speed and GPU Memory Efficiency. Our method demonstrates high speed and less memory usage. We evaluate the floating-point operations (FLOPs), throughput, and

GPU memory consumption in inference on three input feature sizes: 14×14 , 56×56 , and 200×200 . First, we compare three CUDA-based scanning operators: the CUB 1D scan used by Mamba, the naive 2D scan introduced in Section 3.4, and our optimized 2D scan, across the three input sizes with 16 independent state dimensions. As shown in Table 3, our 2D scan significantly outperforms the naive 2D scan in both throughput and GPU memory efficiency across all input sizes, with the performance gap widening as the feature size increases. Our 2D scan matches the throughput of Mamba’s CUB scan for the 14×14 input size. However, as input size increases, its throughput declines compared to the CUB scan. This is due to more complex memory layout of 2D data and our doubled computations. Nonetheless, our 2D scan maintains linear memory consumption with respect to the sequence length. We then assess the CUDA implementation of Mamba, the Python implementation of our 2DMamba, and the CUDA implementation of our 2DMamba within the MIL framework, across the three input feature sizes. Table 3 indicates that our CUDA-based 2DMambaMIL framework consistently outperforms the Python-based implementation across all metrics, benefiting from our hardware-aware 2D scan operator. The throughput of our method remains at 70%-90% of the vanilla Mamba-based MIL framework.

Natural Image Classification. Besides its effectiveness and efficiency on pathology images, our method also generalizes well on natural image classifications. We apply our 2DMamba to the SOTA Mamba-based method on natural images, VMamba [26]. We replace its Mamba block with our 2DMamba block and name it as 2DVMamba. We first evaluate it on the ImageNet-1K classification dataset and compare them with: Swin Transformer [27], Vim [49], EfficientVMamba [36], LocalVMamba [20] and the original VMamba. Table 4 shows that our 2DVMamba achieves 0.2% higher accuracy than the original VMamba and surpasses all SOTA methods.

Natural Image Segmentation. We further evaluate the

Scope	Feature size Method	14 × 14			56 × 56			200 × 200		
		FLOPs	Thro.	GPU Mem.	FLOPs	Thro.	GPU Mem.	FLOPs	Thro.	GPU Mem.
CUDA operator	CUB 1D scan	9K	49K	1.6KB	150K	12K	25.1KB	1.9M	3K	0.3MB
	Naive 2D scan	16K	0.2K	14.1KB	251K	0.06K	225.8KB	3.2M	0.02K	2.9MB
	Our 2D scan	16K	40K	1.6KB	251K	6K	38.5KB	3.2M	1K	0.5MB
MIL framework	Mamba (CUDA)	58M	894	24MB	0.9G	752	58MB	11.8G	203	500MB
	2DMamba (Python)	63M	327	46MB	1.0G	187	430MB	12.9G	13	5842MB
	2DMamba (CUDA)	63M	655	24MB	1.0G	625	76MB	12.9G	185	598MB

Table 3. Comparison of floating-point operations (FLOPs), throughput (Thro., feature maps per second), and GPU memory consumption during inference. CUDA operators are measured using single dimensional feature input and MIL frameworks are measured using 128 dimensional feature input. The state dimension is set to 16 for all experiments.

Method	#Param	FLOPs	Top-1 Acc%
Swin-T	28M	4.5G	81.3
Vim-S	26M	-	80.3
EfficientVMamba-B	33M	4.0G	81.8
LocalVMamba-T	26M	5.7G	82.7
VMamba-T	30M	4.91G	82.6
2DVMamba-T	30M	4.94G	82.8

Table 4. The top-1 accuracy (%) of our V2DMamba-T on the ImageNet-1K dataset. All images are of size 224 × 224.

performance of 2DVMamba on the ADE20K semantic segmentation dataset. Tab. 5 shows that 2DVMamba-T outperforms the baseline VMamba-T, achieving a gain of 0.7 in single-scale mIoU, 0.5 in multi-scale mIoU and surpassing all baselines. Notably, the enhancement in segmentation performance is more pronounced compared to that in classification. This is likely due to that segmentation is a dense prediction task where maintaining spatial continuity across patches is crucial.

Method	#Param.	FLOPs	mIoU(SS)	mIoU(MS)
Swin-T	60M	945G	44.5	45.8
Vim-S	46M	-	44.9	-
EfficientVMamba-B	65M	930G	46.5	47.3
LocalVMamba-T	57M	970G	47.9	49.1
VMamba-T	62M	949G	47.9	48.8
2DVMamba-T	62M	950G	48.6	49.3

Table 5. The performance of our 2DVMamba-T on the ADE20K semantic segmentation dataset. “SS” and “MS” denote single-scale and multi-scale testing, respectively. FLOPs are calculated with an input size of 512 × 2048.

Ablation on the non-tissue padding. We ablate our learnable padding token for non-tissue regions by comparing it with a naive solution: padding all fixed zero tokens, on the PANDA and TCGA-BRCA datasets. Table 6 shows

that our learnable padding outperforms the fixed padding by relatively 1.56%-4.25% and 0.62%-1.58% in accuracy and AUC, respectively. This suggests that our trainable padding enables the scanning to adapt more effectively to the non-tissue regions.

Padding token	PANDA		TCGA-BRCA	
	Acc	AUC	Acc	AUC
Fixed zero	0.4868	0.8057	0.9313	0.9722
Learnable	0.5075	0.8184	0.9458	0.9782

Table 6. Ablation on the non-tissue paddings on the PANDA and TCGA-BRCA dataset. Our learnable token achieves higher performance compared to the fixed zero token.

Ablation on the multi-directional scanning. We ablate MambaMIL using 2-direction [49], 4-direction raster [45], and 4-direction cross scans [26], comparing with 2DMambaMIL. Tab.7 shows that while scanning in multiple directions improves performance, it remains inferior to 2DMambaMIL. This demonstrates that multi-directional scanning does not accurately model cell-to-cell interactions, as cells interact in a coordinated manner across all directions, rather than being limited to horizontal and vertical orientations.

Method	PANDA		TCGA-BRCA	
	Acc	AUC	Acc	AUC
MambaMIL (1D)	0.4679	0.7781	0.9333	0.9657
w. 2-direction	0.4853	0.7749	0.9374	0.9753
w. 4-direction (raster)	0.4923	0.7918	0.9388	0.9755
w. 4-direction (cross)	0.4939	0.8006	0.9402	0.9698
2DMambaMIL (2D)	0.5075	0.8184	0.9458	0.9782

Table 7. The comparison of MambaMIL with 2-direction [49], 4-direction scan [26, 45] and 2DMambaMIL.

Qualitative Evaluation. We qualitatively compare the attention heatmaps generated by 2DMambaMIL with four existing approaches (AB-MIL, CLAM, MambaMIL, and SRMambaMIL) for classification and survival analysis

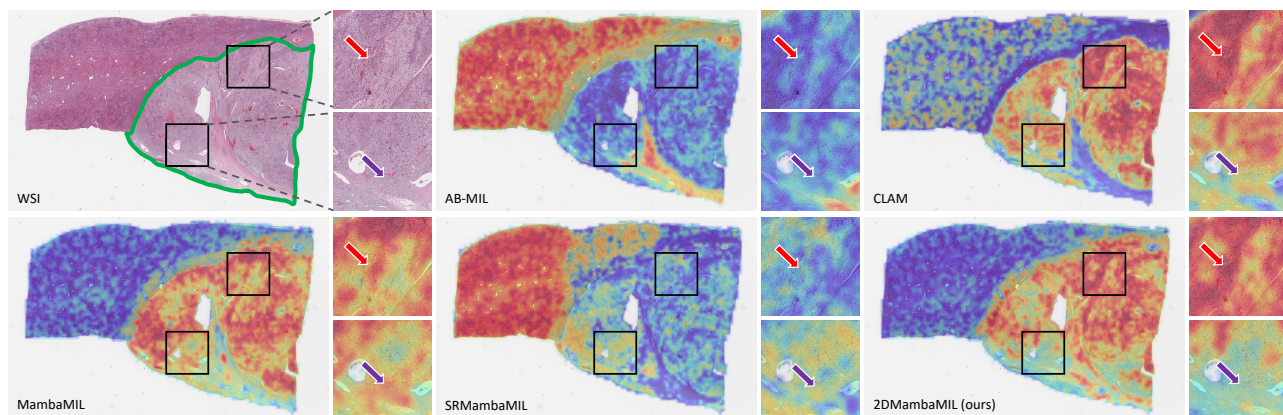


Figure 4. The attention visualization of 2DMambaMIL and four other methods on a TCGA-KIRC sample for survival analysis. Tumor regions are outlined in green. AB-MIL and SRMambaMIL primarily focus on non-tumor areas, while CLAM also shows substantial attention to non-tumor regions. In contrast, both 2DMambaMIL and MambaMIL focus predominantly on tumor regions. Compared with MambaMIL, attention of 2DMambaMIL shows a more heterogeneous fashion, focusing more on critical regions related to survival (red arrows) while paying less attention to less related ones (violet arrows).

tasks, focusing on pathological and biological interpretability. The results demonstrate that 2DMambaMIL consistently targets tumor areas in WSI classification and survival analysis datasets, occasionally including pixels from the immediate tumor-adjacent regions. Fig. 4 presents a case of kidney clear cell carcinoma in the context of survival analysis. AB-MIL and SRMambaMIL predominantly attend to non-tumor regions, which is unnecessary for risk prediction, and CLAM also shows considerable attention to non-tumor areas. On the contrary, the attentions of our 2DMambaMIL and SRMambaMIL are both driven by tumor areas. In comparison, our method exhibits a more heterogeneous attention pattern, specifically focusing on highly survival-related regions (indicated by the red arrows), whereas SRMambaMIL’s attention is more uniformly distributed, which focuses on some less survival-related regions (indicated by the violet arrows). Additional qualitative evaluations are shown in Supp. I.

Visualization of Effective Receptive Fields. Effective Receptive Fields (ERF) refers to the region in the input space that contributes to the activation of a certain output unit [30]. We conduct analyses of the central pixel’s ERF between Swin-T, VMamba-T and 2DVMamba-T. Fig. 5 shows that the ERF of Swin-T expresses a local pattern, consistent with its local structure. VMamba-T expresses a more global pattern but has a clear cross signal resulting from its 4-way 1D scan process. 2DVMamba demonstrates much more global and smooth ERFs without cross-signal showcasing to persevere spatial continuity.

5. Conclusion

In this work, we presented 2DMamba, a novel 2D selective SSM framework incorporating the 2D spatial structure

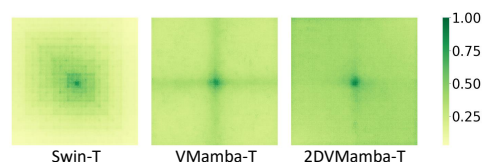


Figure 5. Comparison of Effective Receptive Fields (ERF) between Swin-T, VMamba-T and 2DVMamba-T. Pixels with higher intensity indicate larger responses regarding the central pixel.

of images into the Mamba paradigm. Unlike the vanilla Mamba processing a flattened 1D sequence, 2DMamba employs a 2D selective SSM architecture to capture both geometric and semantic information directly from 2D feature maps using a hardware-optimized selective scan operator. We evaluated 2DMamba on 10 WSI classification and survival analysis datasets, where our method consistently outperformed conventional MIL methods and 1D Mamba-based MIL approaches. Furthermore, our design enhances GPU efficiency through optimized caching and increased parallelism for 2D structures. We also demonstrated its strong generalizability on natural image classification and segmentation tasks by integrating it into a SOTA method. Future work will focus on refining 2D SSM designs and exploring broader applications.

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